

HGC³AE² at the Degraded Edge

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2026-04-30

HGC³AE² AT THE DEGRADED EDGE

Series 3 — The Implications of Edge Degraded Ops | P2 of 11 | v0.1-seed

§1 — REFRAME: HGC³AE² FOR THE SERIES 3 READER

HGC³AE² is a framework — not a product, not a platform, not a configuration profile. It defines *who governs, what gets curated, and what gets enforced* in an agentic system. That ordering is load-bearing: the H half governs; the C³ half curates, catalogs, and teaches; the AE² half authenticates and enforces. AI executes within the envelope that ordering defines. The framework does not specify any particular technology stack, link type, or deployment topology. It specifies a *grammar of governance* — a way of describing who holds authority, over what scope, and how that authority is made legible to a system that will act on it. Readers arriving from Series 1 P1 [@HGC3AE2-S1P1] already hold that grammar. This paper does not re-derive it. It applies it.

Series 3 applies HGC³AE² to a distributed substrate operating under DDIL (Denied, Degraded, Intermittent, and Low-bandwidth) constraint. The substrate is the same one P1 described [@S3P1-TacticalSubstrate]: a layered stack of Delay-Tolerant Networking (DTN) at the transport layer, a write-ahead log (WAL) as the persistence and replication layer, conflict-free replicated data types (CRDTs) as the merge-coherence layer, and an AI Supervisor as the control-plane inference engine. That substrate is built to survive link degradation. What P1 surveyed at architectural width, P2 examines at governance depth: how does the HGC³AE² framework sit on top of that substrate, and what does each half *do* when the link is marginal or absent?

The paper's claim is precise: the HGC³AE² framework does not change in shape at the degraded edge. What changes is the *mechanism* — how each half is implemented given link constraint. A blackout window does not suspend human governance; it reveals which governance decisions must be pre-encoded into the substrate versus which can be deferred to reconnection. A degraded link does not suspend curation; it tightens the curation envelope, forcing prioritization over completeness. A broken authentication chain does not suspend enforcement; enforcement fails closed. The framework travels — unchanged in grammar, specialized in mechanism per layer and per operational mode. This is what “link-agnostic in shape, link-aware in mechanism” means.

The intended reader has completed P1 and holds the Series 3 substrate model. Required prior reading beyond P1 includes HGC³AE² itself [@HGC3AE2-S1P1] and the Operational Discipline Doctrine [@OpDiscipline-S1P2]. P2 is the governor lens through which P3–P11 are focused: every problem-unpack paper in this series applies the framework the P2 sections define. Readers approaching P3–P11 without P2 will find the §6 Governor Application sections opaque. Readers who hold both P1 and P2 will find §6 in each of those papers to be a direct application of a grammar they already speak.

§2 — H HALF: HUMAN GOVERNANCE APPLIED TO THE SUBSTRATE

Human governance, in the HGC³AE² framework, is not approval-click governance. It is not a human being asked to ratify every packet transit or every AI inference cycle. It is the set of decisions that are *structurally reserved for human judgment* — decisions whose delegation to an AI Supervisor would constitute a category error under the Operational Discipline Doctrine [OpDiscipline-S1P2]. On the Series 3 substrate, five classes of decision belong unambiguously to the H half.

The first is drift threshold governance. When a ship-authoritative node and a shore-authoritative node diverge in state, the delta between them is a measurable quantity. At some delta, the divergence is operationally acceptable — the system tolerates it and will reconcile at reconnection. At some larger delta, the divergence is operationally dangerous — the AI Supervisor must trigger a forced sync or flag the condition for human action. Where that threshold sits is a governance decision. It is not an engineering default that ships with a reasonable value. It is a decision that carries operational consequence, made by the humans who understand the operational envelope, encoded into the substrate as a policy parameter the AI Supervisor enforces but does not set.

The second is classification routing policy. Which data crosses which boundary under which conditions is a human-governance decision at its root. The Series 3 substrate carries data of multiple classification levels across a network topology where node accreditation varies. Which data moves in degraded mode, which is held for Connected mode, and which is never transmitted on a given link — these are policy choices that humans make. The AI Supervisor enforces the routing policy; it does not invent it. A system that delegates this decision to the AI Supervisor has inverted the dependency chain — enforcement before governance is the failure mode the Operational Discipline Doctrine names explicitly.

The third is blackout mode trigger conditions. The transition from Degraded mode to Blackout mode is a state change with significant operational consequence: AI-autonomous behavior narrows, curation becomes more aggressive, and certain data flows halt. The conditions under which that transition fires — and crucially, who holds authority to override the AI Supervisor's recommendation when those conditions are borderline — are governance decisions. The AI Supervisor may observe and recommend; it does not decide. Human authority over the blackout trigger is not an edge case provision. It is the load-bearing constraint that keeps the AI Supervisor in its autonomy envelope.

The fourth is conflict arbitration policy. When two authoritative copies of the same state object diverge beyond the CRDT merge window — when the system cannot automatically resolve the conflict because the operational semantics of the conflict require judgment the substrate cannot encode — a human must arbitrate. Who that human is, by what process the conflict is surfaced, and within what time window a decision is expected: these are governance artifacts. The substrate can propose a resolution; it cannot decide one. An AI Supervisor that decides conflicts without human arbitration has exceeded its autonomy envelope, and the AE² enforcement mechanisms exist precisely to prevent that exceedance.

The fifth is the AI Supervisor's own boundary. The scope of AI autonomy — what the Supervisor may do, in what modes, on what data classes, without human confirmation — is itself a human-governance artifact. The red line (the AI Supervisor stays in the control plane; it does not take actions in the data plane without explicit human authorization) is set by humans and enforced by AE². It is not a preference or a default. It is the outer wall of the governance envelope, and it must be defined before the substrate is deployed. The H half is, in this sense, prior to the substrate: governance decisions are made before the system runs, encoded as policy parameters, and enforced by the substrate during operation. The human who is

unreachable during a blackout window still governs — because the decisions they made before the blackout are running in the substrate as policy.

§3 — C³ HALF: CURATION, CATALOGING, AND CURRICULUM APPLIED TO THE SUBSTRATE

The C³ half of HGC³AE² names three distinct but interlinked functions: Curation (deciding what matters and what gets priority), Cataloging (recording what happened with enough fidelity that the system knows its own state), and Curriculum (building the AI Supervisor's operational model from cataloged history). On the Series 3 substrate, each function has a direct architectural implementation, and the three functions compose into a capability the substrate cannot have if any one is missing: a system with memory and judgment operating inside a governance envelope.

Curation is the function of deciding, before the link constraint forces the decision, which state is worth transmitting and in what order. Not all state is equally important. In degraded mode, the AI Supervisor cannot transmit everything — bandwidth is scarce, link windows are unpredictable, and the cost of a failed custody transfer is the state that doesn't arrive. The curation policy specifies how state is classified by operational criticality, recency, and classification level — and then how those classifications translate into transmission priority under each operational mode. A targeting update is not the same as a logistics log entry; a sensor reading from the last ten minutes is not the same as a sensor reading from last week. The curation policy is human-set; the AI Supervisor executes it under constraint. What the AI Supervisor may not do is override the curation policy in real time based on its own inference about operational importance — that path leads back to the AI-execution-before-human-governance failure the framework exists to prevent.

Cataloging is the function the write-ahead log performs. Every write across the distributed substrate is a catalog entry: timestamped, versioned, tagged with its classification level, ordered for custody transfer, and cryptographically signed by the originating node. The WAL is not a backup mechanism or a recovery tool — it is the ground truth of what happened, in what order, on which node. When a ship-authoritative node and a shore-authoritative node reconnect after a blackout window, the first question the system asks is: what does each catalog contain, and what is the delta? The WAL answers that question with precision, because every write was cataloged in order. Without the WAL, the substrate does not know what it knows. Cataloging is how the system achieves the situational awareness the H half's governance decisions require.

Curriculum is the function that builds the AI Supervisor's operational model from cataloged history. The Supervisor is not a static inference engine — it learns from operational cycles. Which link states predict which failure modes? Which drift patterns resolve cleanly at reconnection versus which require human arbitration? Which custody chains complete within acceptable time windows under which RF conditions? The answers to these questions are not derivable from first principles at deployment time. They emerge from operational history — from the catalog the WAL maintains. The Supervisor's model of the operational envelope, its ability to distinguish a recoverable degradation from an escalating failure, its calibration of drift thresholds against actual operational outcomes — all of it is curriculum, built from cataloged history and updated with each operational cycle. The Supervisor that has accumulated curriculum is a qualitatively more capable control-plane engine than the Supervisor at initial deployment, because it has learned the envelope it is operating in.

The three functions compose. Curation sets what matters — which state, in this operational context, is worth the custody chain cost. Cataloging records what happened — in what order, on

which node, with what classification, with what cryptographic integrity. Curriculum teaches the AI Supervisor to predict what will happen — which operational signals precede which outcomes, which governance parameters are binding in which conditions. Together, they give the substrate a memory (the WAL) and a judgment (the Supervisor's trained model) operating inside a governance envelope (defined by the H half) and enforced at the boundary (by AE²). Remove any one function and the substrate degrades: without curation, the AI Supervisor is prioritizing by mechanism rather than by operational meaning; without cataloging, the system cannot know its own state delta; without curriculum, the Supervisor is flying blind at every blackout exit.

The load-bearing insight of the C³ half for Series 3 is that these functions must be designed *into* the substrate before degraded operation begins. Curation policy that is written during a blackout window is too late. A WAL that was not structured for custody transfer before the link degraded cannot be restructured while it is degraded. A Supervisor with no operational curriculum at first deployment is a governance liability — it will make inference errors on an envelope it has not yet learned. Series 3's nine problem papers (P3–P11) each carry a §6 subsection on C³ that applies these three functions to the specific problem that paper addresses.

§4 — AE² HALF: AUTHENTICATION AND ENFORCEMENT APPLIED TO THE SUBSTRATE

The AE² half of HGC³AE² names two interlocked functions: Authentication (every boundary crossing is cryptographically authenticated) and Enforcement (every rule is enforced fail-closed). On the Series 3 substrate, these functions are not security overlays — they are architectural load-bearing elements. A substrate that operates without them is not a secured substrate with a gap; it is a different substrate with a different trust model. AE² is designed into the substrate, not applied to it after the fact.

Authentication on the Series 3 substrate covers every boundary crossing without exception. DTN bundles are signed at the custody source and verified at each custody hop: a bundle with a broken signature chain does not traverse the substrate, regardless of the operational urgency of its payload. Every inter-node connection runs mutual TLS (mTLS) — both sides of every connection present credentials; neither side trusts the other on assertion alone. Every data object carries its classification tag, signed by the originating node's PKI credential. Classification tags are not advisory labels — they are enforced at every Cross-Domain Solution (CDS) crossing, and a tag that cannot be cryptographically verified is treated as an unclassified object by default (fail-closed, not fail-open). The AI Supervisor's control-plane commands are authenticated by the same mechanism as data-plane writes: a spoofed control command does not execute, because it carries no valid credential. The Skipjack Protocol [@Skipjack-S1Cap] defines the operational discipline frame within which these authentication mechanisms operate — Series 3 inherits that frame and applies it at substrate scale.

Enforcement on the Series 3 substrate is fail-closed everywhere. This is not a posture choice or a risk tolerance calibration — it is a structural commitment. The CDS placement enforces mechanically that unclassified data does not cross a classified boundary without explicit CDS passage: the CDS is not a policy advisory that can be overridden in the field; it is an architectural component that makes the crossing physically impossible without proper processing. Multi-Level Security (MLS) rules govern the WAL replication path: TS data does not replicate to a node without TS accreditation, regardless of link state or operational tempo. FIPS-validated cryptography runs on every path the substrate uses. The substrate does not offer a

non-FIPS path as a fallback, a legacy accommodation, or a performance optimization — there is no non-FIPS path. Every path is FIPS-validated or the path does not exist.

The kill-switch provision is an AE² mechanism that deserves explicit treatment. The AI Supervisor has an explicit kill-switch: a human-accessible control, implemented at the hardware level where possible, that suspends all AI-autonomous behavior and returns the system to manual operation. This is not an emergency stop for safety theater — it is the AE² mechanism that enforces the outer boundary of AI autonomy. When the kill-switch fires, the Supervisor's autonomy envelope collapses to zero: no autonomous custody transfers, no autonomous routing decisions, no autonomous drift-threshold enforcement. A human operator takes direct control. The kill-switch is the physical embodiment of the governance principle that the H half established: the AI Supervisor operates within a human-defined envelope, and that envelope can be withdrawn at any time, by a human, without the Supervisor's consent. The Edge AI Doctrine [EdgeAIDoctrine-S1P3] establishes the inference-at-edge governance framework from which this provision is inherited; Series 3 extends it to state-sync-at-edge.

The ordering of AE² relative to H and C³ is not symmetrical, and the asymmetry is load-bearing. AE² enforces what the H half has decided and what the C³ half has cataloged. Enforcement without governance is mechanism without purpose: AE² can enforce a rule it does not understand, but the rule it enforces must come from somewhere, and if it does not come from H, it comes from an engineering default — which is, structurally, AI-defined governance. Enforcement without curation enforces the wrong thing: a CDS that is enforcing classification tags on data that was mis-curated at origin is enforcing a fiction. The dependency ordering (H → C³ → AE²) runs through the entire framework, and the AE² half is the last term in the chain — the most mechanically precise, and the most dependent on the two halves that precede it.

§5 — THE INTEGRATED PICTURE

Figure 1 — HGC³AE² Governor Structure Across the Tactical Substrate *Forthcoming v1.0-preprint: substrate diagram (from P1 Figure 1) with H / C³ / AE² responsibilities color-coded across every component — DTN custody chain (AE²), WAL cataloging (C³), AI Supervisor control plane (C³ + AE²), human governance boundaries (H).*

[FIGURE PLACEHOLDER — net-new diagram required; no v4 deck slide covers this. Commission at v1.0-preprint polish. Base: slide-02 (Operational Picture) or slide-09 (AI Supervisor) with HGC³AE² overlay.]

The integrated picture shows the Series 3 substrate from P1 with the governor structure superimposed. The DTN bundle custody chain is an AE² mechanism. The WAL is the C³ catalog. The AI Supervisor operates at the intersection of C³ (curriculum-informed inference) and AE² (enforcement of the control-plane boundary). The H half is visible as the governance envelope that defines the scope of all three. No component of the substrate operates outside the governor structure — this is what “link-agnostic in shape” means: the substrate components change; the governance grammar does not.

§6 — MANDATORY-SECTION CONTRACT: THE §6 TEMPLATE FOR P3–P11

Every paper in Series 3 from P3 through P11 carries a §6 Governor Application section. That section is not optional, not a summary of this paper, and not a boilerplate insert. It is a substantive application of the HGC³AE² framework to the specific problem that paper addresses

— at the scale and depth that problem requires. This section defines the contract that §6 in every problem-unpack paper must satisfy.

The contract exists because the nine problem papers (P3–P11) are not independent documents. They are applications of a shared framework to distinct problem spaces. A reader who moves through P3, P4, and P5 in sequence should be able to read the §6 sections as a coherent set: the same grammar, applied consistently, at different problem scales. That consistency is a quality guarantee for the series. It is also an IP-race provision: if any problem paper's §6 is underspecified, the governance application for that problem is incomplete, and the paper cannot stand as a credible technical claim. The contract below is the minimum specification for a complete §6.

The four-subsection structure (§6.1 H, §6.2 C³, §6.3 AE², §6.4 Reference to P2) is mandatory. Each subsection addresses the corresponding governor half of HGC³AE² at the scale of the paper's specific problem. §6.1 names the human governance decisions that are load-bearing in this problem space — not all governance decisions, but the ones where a wrong delegation produces a failure mode the architecture cannot recover from. §6.2 names the curation, cataloging, and curriculum functions as they apply to this problem's specific state and operational patterns. §6.3 names the authentication boundaries and fail-closed enforcement rules specific to this problem, including the kill-switch provision for this problem's AI-autonomous behavior. §6.4 closes with an explicit reference to the relevant P2 section — anchoring the problem-specific treatment to the substrate-wide treatment this paper provides.

Length and depth expectations for §6 are explicit: approximately 400–600 words across all four subsections. This is not a short note or a bullet-point summary. It is a substantive application of the framework that a competent reader could use to evaluate whether the paper's governance claims are complete. A §6 that runs shorter than 400 words has almost certainly underspecified at least one subsection. A §6 that runs longer than 600 words in a problem-unpack paper is probably duplicating P2 rather than applying it — the test is whether the words are doing problem-specific work or restating the framework at length.

The §6 template follows. It is a contract, not a suggestion. Every P3–P11 paper implements this template at its specific problem scale.

§6 Governor Application — Mandatory Contract for Series 3 P3–P11

§6 applies HGC³AE² to the specific problem set addressed in this paper. It is not a summary of P2. It is an application of P2's framework at problem-set scale.

§6.1 — H: Human Governance

What decisions, in this problem space, require human judgment? What decisions are pre-delegated to the AI Supervisor's autonomy envelope? Where is the governance boundary load-bearing — i.e., where does a wrong delegation produce a failure mode that the architecture cannot recover from?

§6.2 — C³: Curation, Cataloging, Curriculum

What state, in this problem space, requires curation (prioritization, classification, not just storage)? What does the catalog (WAL) record that is specific to this problem? What does the AI Supervisor learn — what patterns, what failure modes, what operational signals — from the history of this problem?

§6.3 — AE²: Authentication and Enforcement

What boundaries, in this problem space, require authentication? What rules are enforced fail-closed? Where is the kill-switch provision for this problem's AI-autonomous behavior?

§6.4 — Reference to P2

Close §6 with: “See P2 — ‘HGC³AE² at the Degraded Edge’ §{relevant section} for the substrate-wide treatment of [the specific governor axis most load-bearing in this problem]. This section applies that treatment at the scale of [this problem].”

Length: §6 should be ~400–600 words across all four subsections for a problem-unpack paper. It is not a short note — it is a substantive application of the framework.

The subsection ordering ($H \rightarrow C^3 \rightarrow AE^2$) is not arbitrary. It mirrors the dependency ordering: governance defines the envelope; curation operates inside it; enforcement secures the boundary. An implementation that enforces before it has curated, or curates before it has governance, is inverting the dependency chain — and will produce the category-error the Operational Discipline Doctrine [@OpDiscipline-S1P2] names as the AI-execution-before-human-governance failure.

This ordering is enforced by the template, not merely recommended. A §6.3 that identifies enforcement rules that exceed the governance scope defined in §6.1 is internally inconsistent. A §6.2 that curates state the governance in §6.1 has not authorized is a curation violation. The template structure forces internal consistency — if the writer cannot specify §6.1 and then find consistent material for §6.2 and §6.3, the problem space itself has a governance gap, and that gap is the finding the paper should surface, not paper over.

BIBLIOGRAPHY

Citation keys from assets/series-3-edo/series-3-citations.bib.

Load-bearing references (□□□)

[@HGC3AE2-S1P1] Yeti. “HGC³AE² — Human Governance, Curation³, and AI Enforcement² as an Integrated Framework for Agentic Operations.” *nonsequitur.tech*, 2025. [NSQ-PUBLISHED] Series 1 P1. Load-bearing parent for all Series 3 papers; the $H / C^3 / AE^2$ split is the engineering grammar of every Series 3 deliverable.

[@Skipjack-S1Cap] Yeti. “The Skipjack Protocol — Operational Discipline for Agentic Systems Under Constraint.” *nonsequitur.tech*, 2025. [NSQ-PUBLISHED] Series 1 capstone. Operational discipline frame; Series 3 lives inside Skipjack.

[@OpDiscipline-S1P2] Yeti. “Operational Discipline Doctrine — Five Deliberate-Stop Classes for Human-Governed Agentic Systems.” *nonsequitur.tech*, 2025. [NSQ-PUBLISHED] Series 1 P2. Five deliberate-stop classes; load-bearing for all §6 Governor Application sections.

Supporting references (□□)

[@EdgeAIDoctrine-S1P3] Yeti. “Edge AI Doctrine — Inference at the Tactical Edge: Governance, Constraint, and the Limits of Autonomy.” *nonsequitur.tech*, 2025. [NSQ-PUBLISHED] Series 1 P3. Direct predecessor to Series 3; extends inference-at-edge to state-sync-at-edge.

[@S3P1-TacticalSubstrate] Yeti. “The Tactical Substrate.” *nonsequitur.tech*, Series 3: The Implications of Edge Degraded Ops, P1, 2026. v0.1-seed; DOI pending Zenodo deposit. Required prior reading.

Forthcoming — Series 3 P3–P11 (listed as forthcoming)

P3 — The Transport Problem (*forthcoming*)

P4 — The State Coherence Problem (*forthcoming*)

P5 — The Orchestration Problem (*forthcoming*)

P6 — The Protocol Selection Problem (*forthcoming*)

P7 — The Workload Class Problem (*forthcoming*)

P8 — The Classification Problem (*forthcoming*)

P9 — The Accreditation Problem (*forthcoming*)

P10 — The Interoperability Problem (*forthcoming*)

P11 — The Deployment Problem (*forthcoming*)

v0.1-seed — IP race posture. §1–§4 full draft; §5 figure placeholder; §6 mandatory-section contract full draft. Stamp and full bibliography cross-check deferred to v1.0-preprint pipeline.